

- 2 An invasive alien species is a species that has been introduced into an ecosystem where it is not normally found and causes harm to this ecosystem. Invasive alien species can change habitats, reduce biodiversity and cause the extinction of native species. Saltcedar trees, *Tamarix* spp., are invasive alien species introduced to North America from Europe and Asia in the early 1800s.

A biologist carried out an investigation in a woodland ecosystem next to the Virgin River in Arizona, North America. In the woodland, there were areas that contained only saltcedar trees and areas that contained a mixture of native tree species and saltcedar trees.

The biologist wanted to test the hypothesis that:

The diversity of rodents (mice and rats) is lower in areas with only saltcedar trees than in areas with a mixture of native tree species and saltcedar trees.

- (a) Identify the **independent** variable in this investigation.

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..... [1]

- (b) The biologist decided to trap rodents in an area with only saltcedar trees and to trap rodents in an area with a mixture of native tree species and saltcedar trees.

The biologist:

- placed 25 small mammal traps at random sites in each area of woodland
- baited each trap with 10g of food for rodents
- checked each trap after 24 hours
- identified any rodents caught in the trap
- marked the rodents with an ear tag and released them back into the woodland
- carried out this trapping process on four days in April.

The number of **different** individuals of each rodent species trapped during the four trapping sessions was recorded.

- (i) Identify **three** variables, **other than** the small mammal trap and ear tag, that the biologist standardised in this investigation.

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(ii) Describe a method that the biologist could have used to select the trap sites randomly.

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..... [2]

(c) Table 2.1 shows the results of the investigation.

Table 2.1

species	number of individuals (n)	
	area with only saltcedar trees	area with a mixture of native tree species and saltcedar trees
deer mouse	5	4
western harvest mouse	2	1
desert woodrat	0	1
long-tailed pocket mouse	1	2
Merriam's kangaroo rat	0	1
total (N)	8	9

Simpson's index of diversity (D) is a method of assessing biodiversity.

The formula for Simpson's index of diversity is:

$$D = 1 - \left(\sum \left(\frac{n}{N} \right)^2 \right)$$

key to symbols:

n = number of individuals of each species present in the sample

N = the total number of all individuals of all species present in the sample

- (i) The Simpson's index of diversity (D) for the area with only saltcedar trees is 0.531.

Calculate Simpson's index of diversity (D) for the area with a mixture of native tree species and saltcedar trees.

You may use Table 2.2 for your working.

Write the value for Simpson's index of diversity (D) on the dotted line.

Give your answer to three significant figures.

Table 2.2

species	number of individuals (n)	$\frac{n}{N}$	$\left(\frac{n}{N}\right)^2$
deer mouse	4		
western harvest mouse	1		
desert woodrat	1		
long-tailed pocket mouse	2		
Merriam's kangaroo rat	1		
total (N)	9		

Simpson's index of diversity (D) = [2]

- (ii) With reference to Table 2.1 and the values for D in the two different areas, state **and** explain the conclusions that can be made from the results of the investigation.

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(d) Fig. 2.1 shows a deer mouse, *Peromyscus maniculatus*, from North America.



Fig. 2.1

The biologist extended the study to investigate the abundance of the deer mouse. The biologist wanted to test a new hypothesis that:

The abundance of the deer mouse is higher in areas with only saltcedar trees than in areas with a mixture of native tree species and saltcedar trees.

The biologist used the trapping procedure described previously to calculate the abundance of the deer mouse in eight areas with only saltcedar trees and eight areas with a mixture of native tree species and saltcedar trees.

The biologist then calculated the mean abundance of the deer mouse in each type of woodland.

Fig. 2.2 shows the biologist's results.

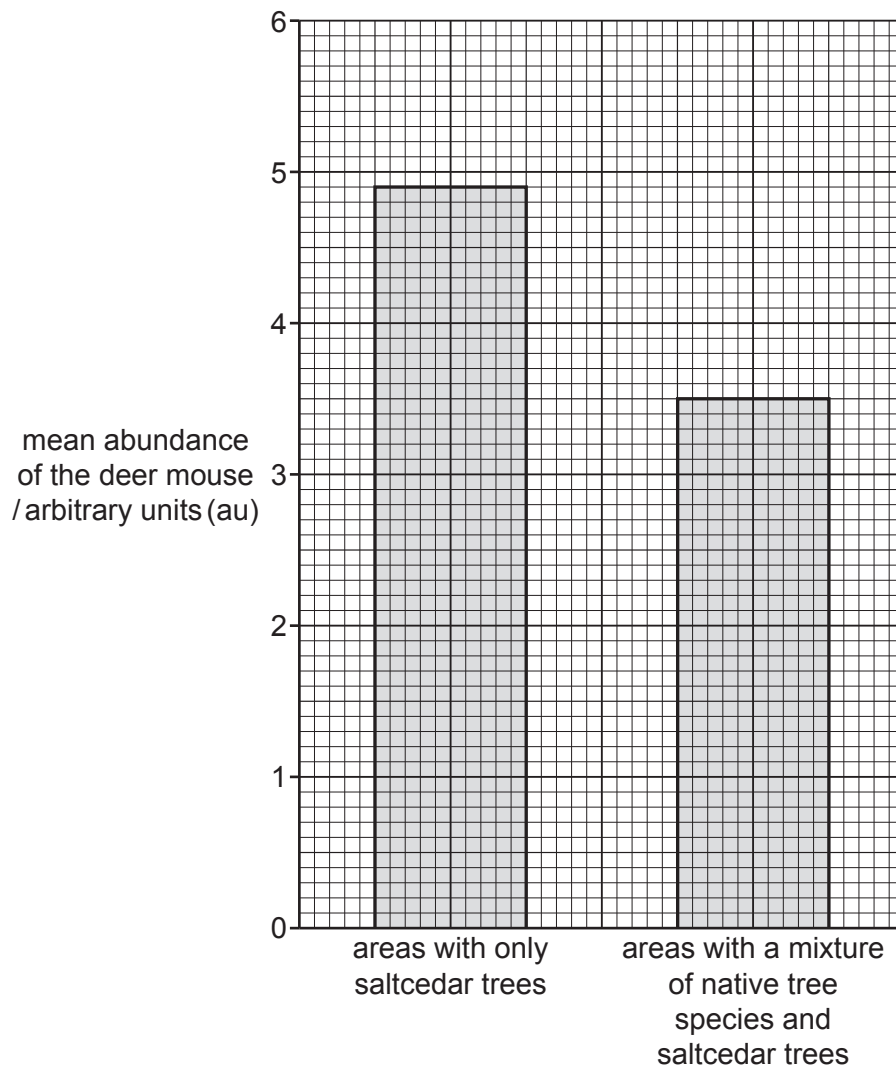


Fig. 2.2

- (i) The standard error (SE) for the mean abundance of the deer mouse in each type of woodland was 0.8 au.

Plot SE error bars on Fig. 2.2.

[1]

- (ii) Explain, with reference to Fig. 2.2, what these SE error bars indicate about the data.

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..... [2]

- (iii) The biologist then analysed these data using a t -test to compare the abundance of the deer mouse in the two types of woodland.

State a null hypothesis for the t -test.

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[Total: 13]